Format Conversion
-Combine FASTA
-EMBL to FASTA
-EMBL Feature Extractor
-EMBL Trans Extractor
-Filter DNA
-Filter Protein
-GenBank to FASTA
-GenBank Feature Extractor
-GenBank Trans Extractor
-One to Three
-Range Extractor DNA
-Range Extractor Protein
-Reverse Complement
-Split Codons
-Split FASTA
-Three to One
-Window Extractor DNA
-Window Extractor Protein
Sequence Analysis
-Codon Plot
-Codon Usage
-CpG Islands
-DNA Molecular Weight
-DNA Pattern Find
-DNA Stats
-Fuzzy Search DNA
-Fuzzy Search Protein
-Ident and Sim
-Multi Rev Trans
-Mutate for Digest
-ORF Finder
-Pairwise Align Codons
-Pairwise Align DNA
-Pairwise Align Protein
-PCR Primer Stats
-PCR Products
-Protein GRAVY
-Protein Isoelectric Point
-Protein Molecular Weight
-Protein Pattern Find
-Protein Stats
-Restriction Digest
-Restriction Summary
-Reverse Translate
-Translate
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-Color Align Properties
-Group DNA
-Group Protein
-Primer Map
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-Translation Map
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-Mutate DNA
-Mutate Protein
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-Random DNA Sequence
-Random DNA Regions
-Random Protein Sequence
-Random Protein Regions
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-Shuffle DNA
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-Home
-IUPAC codes
-Genetic codes
-Browser compatibility
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-About this site
-Acknowledgments
-Reference

Sequence Manipulation Suite:

DNA Stats

DNA Stats returns the number of occurrences of each residue in the sequence you enter. Percentage totals are also given for each residue, and for certain groups of residues, allowing you to quickly compare the results obtained for different sequences.

Paste the raw sequence or one or more FASTA sequences into the text area below. Input limit is 500,000,000 characters.

```
GCAGCCTAAACTGAAACTGGTTATCATGTGAAAGTTCCCTGAACTCAGGCATTCGGGCCTTA
GTCGACCTGGGTAGTCAGCTTCCCGCGCTCTCCGAGATCGTTCGATGTAGGAGCAGCAGTTG
CGGTATTTTAGATCAGGCTCTCGGTCGCACATCAGGGGGTTGTTCCCCGCCATGGGAAGTCG
TCAATAAGGTTACTTAAGGTTAGGAGCGTATACAAAATCAATGATAAACATACGTCTGACAC
CGCCAG
```

Submit | Clear | Reset

*This page requires JavaScript. See browser co
*You can mirror this page or use it off-line.

Sun 1 Oct 12:00:06 2023
Valid XHTML 1.0; Valid CSS

Sequence Manipulation Suite

about:blank

DNA Stats results

Results for 874 residue sequence "sample sequence" starting "GTCTACAGCA"

Pattern:	Times found:	Percentage:
g	225	25.74
a	231	26.43
t	226	25.86
c	192	21.97
n	0	0.00
u	0	0.00
r	0	0.00
y	0	0.00
s	0	0.00
w	0	0.00
k	0	0.00
m	0	0.00
b	0	0.00
d	0	0.00
h	0	0.00

Species
Plants (474)
Customize ...

Molecule types
genomic DNA/RNA (473)
mRNA (1)
Customize ...

Source databases
INSDC (GenBank) (474)
Customize ...

Sequence Type
Nucleotide (474)

Genetic compartments
Chloroplast (471)
Plastid (471)

Sequence length
Custom range...

Release date
Custom range...

Revision date
Custom range...

Clear all

Show additional filters

Summary 20 per page Sort by Default order Send to:

Items: 1 to 20 of 474

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1. **Cistus** ladanifer subsp. ladanifer voucher 01EN09 tRNA-Lys (trnK) gene, partial sequence; and maturase K (matK) gene, partial cds; chloroplast
1,310 bp linear DNA
Accession: HM773247.1 GI: 307645739
Protein Taxonomy
GenBank FASTA Graphics

2. **Cistus** albidus clone 5 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900434.1 GI: 339716237
Protein Taxonomy
GenBank FASTA Graphics

3. **Cistus** albidus clone 4 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900433.1 GI: 339716235
Protein Taxonomy
GenBank FASTA Graphics

4. **Cistus** albidus clone 3 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
349 bp linear DNA
Accession: JF900432.1 GI: 339716233
Protein Taxonomy
GenBank FASTA Graphics

5. **Cistus** albidus clone 2 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900431.1 GI: 339716231
Protein Taxonomy
GenBank FASTA Graphics

6. **Cistus** albidus clone 1 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900430.1 GI: 339716229
Protein Taxonomy
GenBank FASTA Graphics

7. **Cistus** heterophyllus clone 5 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900429.1 GI: 339716227
Protein Taxonomy
GenBank FASTA Graphics

8. **Cistus** heterophyllus clone 4 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900428.1 GI: 339716225
Protein Taxonomy
GenBank FASTA Graphics

9. **Cistus** heterophyllus clone 3 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900427.1 GI: 339716223
Protein Taxonomy
GenBank FASTA Graphics

10. **Cistus** heterophyllus clone 2 RNA polymerase beta-subunit (rpoC1) gene, partial cds; chloroplast
403 bp linear DNA
Accession: JF900426.1 GI: 339716221

Filters: Manage Filters

Results by taxon

Top Organisms [Tree]
Cistus monspeliensis (160)
Cistus salviifolius (116)
Cistus laurifolius subsp. laurifolius (83)
Cistus albidus (22)
Cistus laurifolius subsp. atlanticus (20)
All other taxa (73)
More...

Find related data
Database: Select
Find items

Search details
"Cistus"[Organism] AND 2010/06/04[PDAT] : 2012/02/06[PDAT]
Search
See more...

Recent activity
Turn Off Clear
Cistus[Organism] AND 2010/06/04 : 2012/02/06[PDAT] (474) Nucleotide
Anthoxanthum[Organism] AND 2003/07/25 : 2005/12/27[PDAT] (7) Nucleotide
Anthoxanthum 2003/7/25 2005/12/27 (0) Nucleotide
"Anthoxanthum"[Organism] OR ("Anthoxanthum"[Organism] OR Antl Nucleotide
"Anthoxanthum"[Organism] OR Anthoxanthum[All Fields] (85781) Nucleotide
See more...

```python
# Name: Christopher A. Lee
# Date: 12/30/2025
# Prof: Dr. Callura
# Course: Bioinformatics - ENBC 455

# Problem 3
from Bio import Entrez

Entrez.email = "clee363@terpmail.umd.edu"
handle = Entrez.efetch(db="nucleotide", id=["JX317645", "JQ011276", "NM_001265803", "HM595636", "JQ712981", "JX460804", "NM_013179", "JX491654", "JF927163", "JQ762396"], rettype="fasta")
records = handle.read()
print(records)

# Problem 4
with open('hw2_4.txt', 'r') as file:
    seq = file.readline().strip()
    subseq = file.readline().strip()

    for i, char in enumerate(seq):
        if seq[i: i + len(subseq)] == subseq:
            print(i + 1, "", end="")
```